KO LIH HAN

Software Engineer | Applied AI & LLM Systems

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Software engineer with production backend/full-stack experience and hands-on work building and evaluating LLM systems across Text-to-SQL, hybrid retrieval, and multimodal tool use.

EXPERIENCE

Kahap Inc | Software Developer (Part-time) **Aug 2022 – Aug 2026**

- Primary developer for a production feedback platform, owning submission workflows, automated notifications, and administrative features from implementation through production support.
- Maintained and extended Laravel/Vue.js production applications, shipping business features and debugging application and database issues.
- Handled deployments and server operations, including upgrades, Laravel queues/workers, SSL maintenance, and database troubleshooting.

Pac Com Solution | Full-Stack Developer Intern **Jun 2021 – Jan 2022**

- Developed and maintained full-stack features for production POS applications using Laravel and Vue.js.

PROJECTS

Semantic Text-to-SQL Agent ↗

Python · LangGraph · Qwen3.5 4B · FastAPI · SQLite

- Built a local Text-to-SQL service with deterministic SQL verification, read-only execution, bounded repair, and one-pass clause-guided semantic revision.
- On a frozen 100-query BIRD DEV run, guarded execution raised execution success from 63% to 82%; official BIRD EX moved from 31% to 34%, separating execution reliability from semantic correctness.

Local RAG Search ↗

Python · BM25 · Qwen Embeddings · RRF · FastAPI

- Benchmarked BM25, Qwen dense retrieval, and RRF on 64 official Confluence-compatible queries over 5,189 documents; hybrid reached 0.7630 Recall@10, 0.7418 MRR@20, and 0.8438 Hit@10 with per-result rank provenance.

Hermes Video Tutor ↗

Python · Hermes · Qwen3.5 4B · Tool Calling · Multimodal

- Built a local multimodal video-QA agent that starts with transcript evidence and escalates to frame/clip inspection only when needed; on a frozen 12-question self-authored paired fixture, visual-required full pass reached 66.7% with no unnecessary visual-tool calls on transcript-answerable cases.

SKILLS

AI / LLM: LLM Agents, RAG, Tool Calling, Retrieval & LLM Evaluation, Embeddings, LangGraph

Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js, React

Cloud / Data: AWS, MySQL, Redis, SQLite, Linux

Other: Java, C/C++, Git, pytest | Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | M.S., Information Systems & Applications **2024 – 2026**

GPA 4.09/4.3 | Taiwan | Thesis: *Interpreting Implicit Intent through Workplace Hierarchies and Context*

Chung Hua University | B.S., Computer Science & Information Engineering **2022 – 2024**

GPA 3.9/4.0 | Taiwan